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DEALER BEST PRACTICE SERIES

Road Defect Management Training Exercise

Application Maintenance | Site Management | Component Rebuild | Component Life Management | Safety | MARC Management

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1.0 Introduction

This Best Practice provides the methodology to carry out a training exercise associated with road maintenance (defect management). It is highly recommended that this exercise be included as part of a comprehensive Equipment Management training program.

The exercise consists of a simulation of a hypothetical mine site, comprising a fleet of dump trucks, haul roads, machine repair centers, and management personnel.

After carrying out the exercise, participants will be able to understand the relationship between road conditions, machine deterioration, and the impact proper road management processes have on obtaining the optimal cost per ton.

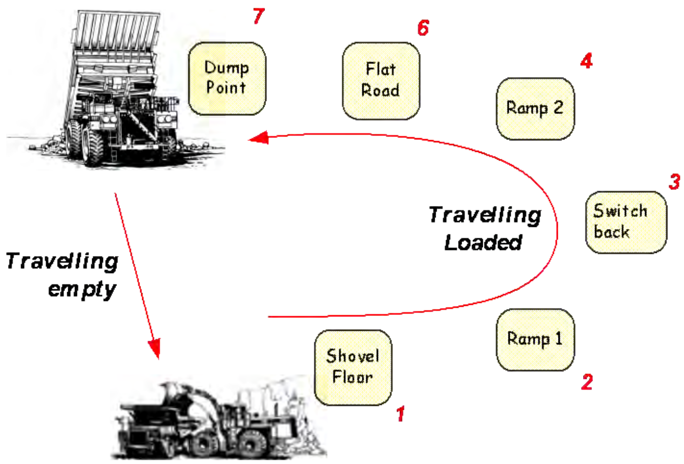

Figure 1 – Road Segments

This Best Practice provides all of the resources required to run this exercise during a training program. The exercise has been designed in a way that any one can edit it and add or remove complexities.

2.0 Best Practice Description

The purpose of this exercise is to provide a top view of the mode of operation of a typical open mine site, and to provide participants an opportunity to understand the relationship between machine application, operation and repair activities. See the attachment **RDM – Exercise Guidelines.doc** for a full description of the training exercise. The main elements of this exercise are (see Figure 2):

Facilitators: Five facilitators are needed for the exercise. Their role descriptions are as follows:

Facilitator #1 – Lead the RDM Exercise:

- Lead exercise round brief, execution, and debrief.
- Act as the "Mine Manager" and set the scene for the group to get the lowest Cost/Ton in the required time frame.

Facilitator #2 – Assist Operations:

- Your main role is to load trucks. Provide the supplied ticket with the payload value to each truck at the beginning at each cycle.
- Provide the "Truck Exchange Time" to each truck coming for a new load, according to the following criteria:
  - 5 minutes: There is another truck in the Shovel Floor at time of loading.
  - 10 minutes: The truck operator dump the load and reload (doesn't like the load).
  - 0 minutes: if none of the two previous conditions is achieved.
- Assisting road segment controllers and machine operators as necessary.

Facilitator #3 – Assist the MARC operation:

- The Repair Shop: Tyres, GET and Power Train.
- FPO and Payload Analysts.

Facilitator #4 – Assist Management:

- Operations Supervisor
- Production Supervisor
- MARC Manager

Facilitator #5 – Capture all data:

- From the exercise and calculate associated metrics in the Facilitator Score Board.
- Capturing Debrief discussion on flipcharts.

Truck Operator: Operate the truck and assess its health condition throughout the exercise.

Wheel Dozer and Motor Grader Operators: objective is to keep Road Segments free of defects.

Haul Road Segment Controllers (Figure 1): objective is to evaluate and document the condition of each road segment according to truck (Payload) traffic and weather (rainfall) conditions.

Production Supervisor: objective is to maximize production at a minimum cost.

Operations Supervisor: objective is to maximize machine availability and utilization (minimum cost).

MARC Manager: on charge of the maintenance and repair activities in the operation, including repair shop mechanics, payload and FPO analysts.

FPO Analyst: objective is to improve cycle time and reduce machine application severity.

Payload Analyst: objective is to improve payload distribution and evaluate the Payload Distribution

Reliability Analyst: objective is to assess machine condition and identify potential problems. Secondary objective is to minimize the number of machine stoppages.

Repair Shop – Tyre Mechanic: objective is to change tyres on machines.

Repair Shop – Power Train Mechanic: objective is to repair power train failures on machines.

Repair Shop – GET Mechanic: objective is to replace GET elements on machines.

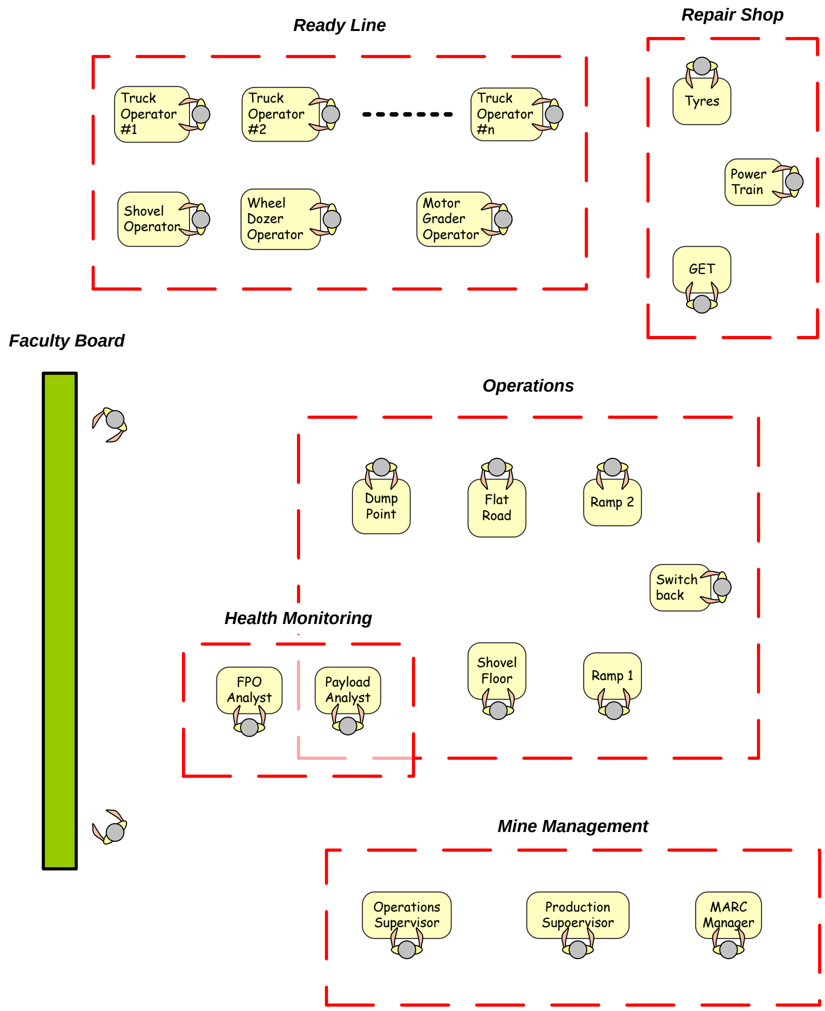

Figure 2 – Exercise and participant layout

Facilitator Score Board

Facilitators must keep the score board updated at all times, by filling in the yellow cells with information from each exercise round. Use a flipchart or a Data Projector to show the Score Board.

Operating Conditions: ***Round 1***

Target Payload:	200 Metric Tons	Rainfall rate:	99 mm
Target Cost/ton:	0.50 $/ton		
Target Truck Availability:	92 %		

Productivity

Cycle	Cycle Time (min)	Production (Metric tons)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
Totals	0	0
Avg.	0.00	0.00

Cost

GET replaced (#)	
Cost per GET ($) 300	0
Tyre exchanges (#)	
Cost per Tyre ($) 20000	0
Power Train repairs (#)	
Cost per repair ($) 50000	0
Total Cost ($)	0
Total Cost/ton	0.00
Total Cost/hour	0.00

Equipment Performance

Total Operating Time (hours)	0.00
(Total cycle time in hours)	
Truck Downtime (min)	
Including: Power Train failures and tyres	
Support Eq. Downtime (min)	
Including: GET changes	
Truck Availability Index (%)	—
Supp. Eq. Availability Index (%)	—

Figure 3 - Facilitator Score Board showing the four areas of analysis

The Facilitator Score Board will display four main areas:

- Operating Conditions:
  - Rainfall Rate
  - Target Payload
  - Target Cost/Ton
  - Contractual Machine Availability
- Productivity Results
  - Cycle times
  - Production values
- Cost Results:
  - Cost per ton
  - Cost per hour
- Equipment Performance Results:
  - Truck Availability
  - Support Equipment Availability

One Score Board must be used at each exercise round.

Starting Conditions
The objective of the exercise is to deliver maximum payload at the minimum cost, over a period of time. The default time of each exercise round is 10 minutes, but the faculty can make it longer if time allows.

The detailed starting conditions are available in the attachment **“RDM – Exercise Guidelines.doc”**

Carrying out the Exercise
The exercise is designed to provide participants an opportunity to understand the relationship between machine application, operation and repair activities. It is also designed to give an opportunity to experience some of the personal aspects leading a project team in a mining environment.

At the end of each round facilitators will debrief and identify some of the personal learning as well as the key learning associated with tools and methodology.

A specific sequence of activities for each round is in the attached file.

Participant Guidelines
The attached file contains job descriptions for each participant, which should be passed out in sealed envelopes before beginning the exercise.

Presenting the Exercise Results: Student’s Wrap-Up Meeting
After finishing all rounds of the exercise, it is recommended to present the results back to the Faculty showing relevant findings that will help to improve their processes and reduce their costs.
This presentation should include preliminary assessment results and identify “low hanging fruit” – items that can be quickly and easily addressed.

Closing discussion
The closing discussion should include preliminary assessment results and identify those “low hanging fruit” items that can be quickly and easily addressed. Selected participants may present the following areas:

Strengths: Start the discussion highlighting what is good and positive as per found from the exercise. Present strong arguments and evidence that supports your findings.

Opportunities: Show those areas where problems are identified. It is advisable to show a viable solution for each problem found in the Operation. Avoid showing problems that may not have any practical solution.

3.0 Implementation Steps

- Recommended number of participants is between 11 and 26.
- Make copies of all required materials, including nameplates and desk papers (see attachments).
- The APD-South Equipment Management team can be contacted for recommendations on carrying out this exercise.

4.0 Benefits

- Serves as a consistent training tool in improving Equipment Management
- Was first conducted during the EM Update (level 2) training program in October 2006 (APD-S, Melbourne)
- Testimonials:
  - o "The Equipment Management group exercise was beneficial to the [training] course."
  - o "Group exercise on Wednesday – good overview of all topics and understand it to the next level."

5.0 Resources Required

Table 1 – Tools and Supplies required during the exercise.

ITEM	Description	Quantity	Comments	
01	Role's hard copies	One per student	Put hard copies on the table before commencing the exercise.	
02	Digital still / video camera	1	Capture the process for future presentation	
03	White board or data projector	1	Faculty will record and display status of the exercise.	
04	Portable memory devices (USB stick)	1	To share information amongst team members	
05	Laser pointer	1	To show potential problems in the white board	
06	A4 envelopes	17	One envelope per role (such as MARC Manager, etc.)	
07	Clipboards	1 per machine	Each Truck, Wheel Dozer and Motor Grader operators record their own information.	
08	Dump Trucks	1 per Truck Operator	http://www.megabloks.com/en/products/description.php?key=1&level=2&level2=1&lId=0&iID=1171&subCat=51	

Refer to RDM Exercise Guidelines.doc

6.0 Supporting Attachments / References

- **"RDM – Exercise Guidelines.doc"** Everything you need to know about the training exercise and how it can be carried out.
- **"Faculty Score Board.xls"** Showing the scor board that can be projected on a screen while conducting the exercise. This board shows metrics associated with cycle times and cost.
- **"A3 Participant Desk Paper.ppt"** Desk papers for participants to use during the exercise. These papers contain correlation tables, such as: road deterioration vs. truck traffic; tyre repair times, etc.
- **"A4 Participant Job and Road Defect Name Plates.ppt"** Name tags for each of the participants and other tickets used during the exercise.
- **"Exercise Results.xls"** The production results of an actual training session.

7.0 Related Best Practices

1106-1.1-1035 Cost Reduction Through Haul Road Management Process.

8.0 Acknowledgements

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